

07b · The same campaign question, on real data — Google’s geo experiment (companion)

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The business decision. A real advertiser spent **\$50,000** on a real ad campaign, in **50 US geos, for four weeks** (16 Feb – 15 Mar 2015), and left 50 geos dark. This is the worked example published with Google’s geo-experiment methodology (Kerman, Wang & Vaver 2017, the `matched_markets` library) — daily **sales and ad cost per geo**, exactly the panel your own company’s data warehouse holds. The question is the one notebook 07 trained us for: **how much incremental sales did the spend cause, and did it pay for itself?**

1.0.1 What changes when the data is real

Notebook 07 planted a +12% lift in a simulator and graded synthetic control against it. Here nobody tells us the truth — that is the whole epistemic situation of applied causal inference. Three things replace the planted truth:

1. **A randomized benchmark.** The 100 geos were split into treatment and control **at random** (in matched pairs, as we’ll verify from the data itself). So a design-based estimate — the matched-pairs difference-in-differences — is *unbiased by construction*. It is noisy, but it is the closest real data ever gets to a known answer, and we grade the synthetic control against it.
2. **Falsification tests** that need no truth: placebo campaigns on untreated geos (Depth B), a fake launch date, and a check the simulator could never give us for free — the campaign *ends*, so the estimated effect must **switch off when the money does**.
3. **Real messiness:** missing geo-days, heavy-tailed market sizes (the largest geo sells 300x the smallest), and a weekly cycle you can see with the naked eye. Each one bites somewhere below, and one of them (market size) breaks a naive estimator in Depth A.

1.0.2 How this notebook is organised

The same **7-step contract** as every notebook — with step 2’s “simulate a ground truth” honestly replaced by “load real data, and say what can still be validated” — plus the anchor depths:

- **Depth A** — the naive estimators on a *single* real market (where they fail for real), and a convex-hull failure the pre-fit gate catches
- **Depth B** — placebo inference: pseudo-cell permutation, the scale-free RMSE-ratio, placebo-in-time

- **Depth C** — the dollar decision and its sensitivity to the launch date, the donor pool, and the margin

FAST=False SC sampling={'draws': 1500, 'tune': 1500, 'chains': 4} pseudo-cells=200

1.1 2 · Load the real data (what replaces “simulate a ground truth”)

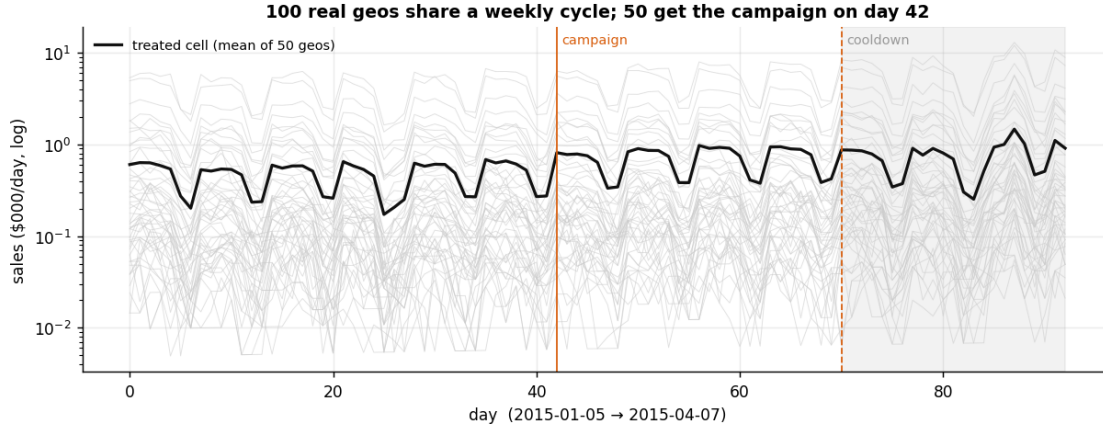
`cmp.data.load_google_geo()` fetches the two public CSVs (Apache-2.0, pinned commit; ~230 KB — cached under `~/.cache/cmp` after the first call) and returns the long panel: `date`, `geo`, `sales`, `cost`, `group`. The **data card**, verified in code below:

fact	value
units	100 US geos, numbered by size (geo 1 = largest — visible in the data)
period	93 days, 2015-01-05 -> 2015-04-07, daily
design	randomized in matched pairs : consecutive size-ranked geos split 1 treated / 1 control
treatment	ad campaign in the 50 group-2 geos, 16 Feb – 15 Mar (28 days), \$50,000 total spend
cooldown	23 days <i>after</i> the spend stops — a free falsification window
blemishes	75 of 9,300 geo-days missing at source (small geos, mostly pre-period) — we interpolate and disclose

Two working conventions. **(a)** We analyse sales in **\$000s** — same numerical scale the fitters were built for. **(b)** The treated unit is the **test cell**: the *mean* daily sales per treated geo. The mean (not the sum) keeps the treated series inside the donors’ range — a sum of 50 geos could never be a convex combination of single geos, and Depth A shows what happens when a series escapes the donor hull.

Eyeball the panel below before any model: every geo rides the same weekly cycle (the shared factors of 07, no longer a modelling assumption but a visible fact), and the treated cell visibly jumps *inside the campaign window only*.

100 geos (50 treated / 50 control) · 93 days · campaign days 42-69 (28 days) ·
 cooldown 23 days
 interpolated geo-days: 75/9300 (0.8%) - small geos, mostly pre-period
 ad spend: \$0 before launch (no anticipation), \$50,000 inside the window, \$0
 after - the design is clean
 geo sizes are heavy-tailed: largest \$6,728/day vs smallest \$18/day (~369x)



1.2 3 · Identify — the estimand, and what randomization buys us

The estimand. Total incremental sales caused by the campaign in the treated geos over the 28-day window:

$$\tau = \sum_{t=T_0}^{T_1} \left(Y_{\text{cell},t}(1) - Y_{\text{cell},t}(0) \right) \times 50,$$

where $Y_{\text{cell},t}$ is mean sales per treated geo and the $\times 50$ restores cell totals. The estimator is exactly 07's: simplex-weighted donors fit to the pre-period fill in the missing $Y_{\text{cell},t}(0)$ (see 07 §3 for the full math and the convex-hull argument). The dollar decision then divides by spend: **IROAS** $= \tau / \$50\text{k}$, incremental revenue per ad dollar.

The assumptions, upgraded by the design. In 07 most rows of this table were unverifiable promises; here randomization converts the worst one into a guarantee:

assumption	status on this real dataset
convex hull / no extrapolation	<i>checked:</i> the treated cell (a mean of 50 geos) sits mid-range of the donors; Depth A shows geo 1 violating it and the gate catching it
good pre-fit	<i>checked:</i> 42 pre-days, PASS/FAIL gate in Step 5
no anticipation	<i>checked from data:</i> \$0 of spend before day 42
stable co-movement	<i>supported:</i> shared weekly cycle visible; placebo-in-time (Depth B) probes it
no spillover between geos	<i>assumed:</i> geos are distinct media markets, but adjacency isn't in the data — the honest caveat that survives even randomization

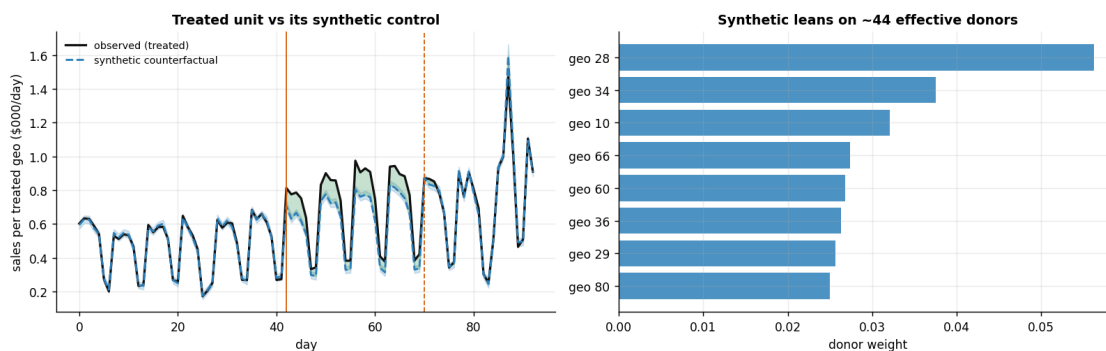
assumption	status on this real dataset
no concurrent treated-only shock	guaranteed in expectation by randomization — a shock would have to coincide with a random coin flip across 50 pairs; in 07 this row was “institutional knowledge only”

Why bother with synthetic control if the design is randomized? Because the design-based estimate, while unbiased, is *noisy* — geo sizes are so heavy-tailed that a plain treatment-vs-control comparison has a wide interval (Step 5 quantifies this). Synthetic control re-weights donors to *match* the treated cell and cuts that variance. And the moment you leave experiments — one treated market, chosen not by coin flip — SC is the only member of this family still standing. That is the real lecture point of this notebook: **the experiment validates the method we’ll use when there is no experiment.**

1.3 4 · Estimate — Bayesian synthetic control on the test cell

Same model as 07: **Dirichlet-simplex weights** over the 50 control geos, fit to the 42 pre-launch days, returning a posterior over the counterfactual path — and therefore over total lift and iROAS. Read three things off the fit, exactly as in 07: **which donors** carry weight (the Dirichlet spreads it — expect a high “effective number of donors” on a mean-of-50 target), the **pre-fit RMSE** (Step 5 turns it into a PASS/FAIL gate), and the **counterfactual band** (a posterior predictive, carrying the cell’s own daily noise, not just weight uncertainty).

SC sampling convergence: max r-hat 1.000 - min ESS 3268 - divergences 56
pre-fit RMSE \$10/geo/day on a ~\$472 level (~2%)
total incremental sales \$149k 90% CI [\$141k, \$158k] on \$50k of spend
revenue iROAS 2.98 90% CI [2.81, 3.15]



1.4 5 · Validate — against the randomized experiment

Four checks. The first is the one no simulator can fake: an *unbiased, design-based* estimate of the same estimand, from the experiment’s own randomization.

1. **The matched-pairs DiD anchor.** The assignment pattern in the data shows consecutive size-ranked geos were randomized one-treated-one-control (a matched-pairs design — we verify this in code). So the mean of the 50 within-pair difference-in-differences is unbiased for the campaign effect, with an honest standard error from the pair scatter. It is our stand-in for 07's planted truth: **the synthetic control point should land inside the anchor's interval.** Note how *wide* the anchor's interval is — unbiasedness isn't precision; this is exactly the variance the model is there to cut.
2. **Pre-fit gate** — same concrete rule as 07: pre-RMSE below a third of the daily lift the design must detect, printed PASS/FAIL.
3. **The off-switch falsification.** The spend stops on day 70; a real advertising effect should collapse toward zero in the 23-day cooldown (up to carryover). A gap that *persists* after the money stops would smell of a confound rather than a campaign. This replaces the simulator's “we know the truth after week 40” with a physical prediction the data can refute.
4. **Leave-one-out donors** — drop each top-weighted donor, refit (fast SLSQP), watch stability.

What we cannot do on real data — and say so plainly, as 07's multi-seed check taught us: we cannot refit on fresh copies of the world, so we cannot *measure* interval calibration here. 07 found SC's total-lift band runs **optimistic** even when the point is right; carry that lesson — trust the placebo permutation (Depth B) for significance, not the band width alone.

anchor: matched-pairs DiD total \$147k +/- 42 (90% CI [78, 216]k; iROAS 2.94 [1.56, 4.32])

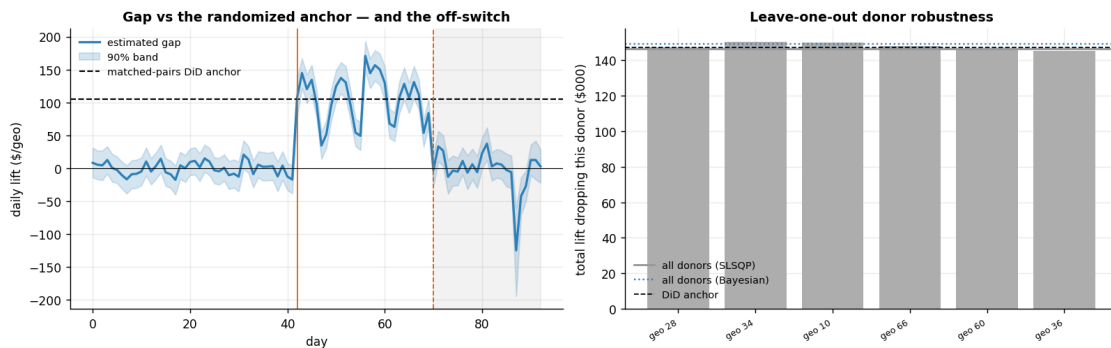
SC total \$149k (iROAS 2.98) - lands on the anchor's point with a ~3x tighter band (the variance reduction SC is for; mind 07's optimism lesson on band width)

pre-fit gate: pre-RMSE \$10 vs |daily lift|/3 = \$36 -> PASS

off-switch: campaign-window gap \$+107/geo/day -> cooldown \$-3/geo/day

(collapses when the spend stops - and no long carryover to book as extra value)

leave-one-out: total ranges \$145-150k vs SLSQP baseline \$147k - no single donor drives it



Read-out. The synthetic control lands almost exactly on the randomized answer — **\$149k vs the anchor's \$147k** (iROAS 2.98 vs 2.94) — passing the only exam real data can set. And the comparison teaches why the model exists at all: the *unbiased* anchor carries a +/- \$70k-wide 90% interval (heavy-tailed geo sizes make even 50 randomized pairs noisy), while SC's band is ~3x tighter around the same point. The gap path shows a clean rectangle: nothing before day 42, a

sustained ~\$105/geo/day lift during the four paid weeks, and a collapse to ~−\$3/geo/day within the cooldown — the effect **switches off when the money does**, which simultaneously validates the attribution and tells the media planner there is no adstock annuity to book. Leave-one-out totals move by ~\$3k on a \$147k estimate: no single market carries the result.

1.5 6 · Decide, in dollars

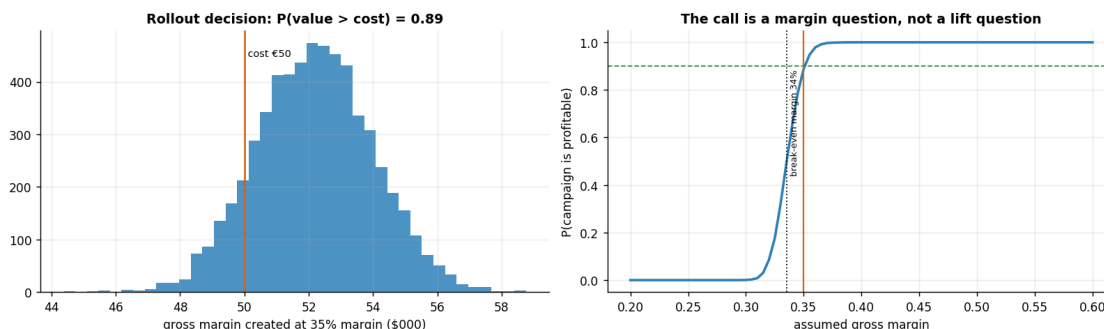
The trap in every iROAS deck: **3.0 revenue-iROAS is not “3x return”**. Ad spend buys *revenue*; the firm keeps only the **gross margin** on it. The campaign pays if

$$\text{iROAS} \times \text{margin} > 1 \quad \Leftrightarrow \quad \text{iROAS} > \frac{1}{\text{margin}},$$

so at our assumed 35% margin the bar is not 1.0 but **2.86** — and our posterior sits almost on top of it. That is the honest shape of this decision: the *lift* is beyond doubt (Depth B), the *profitability* is a knife-edge that turns entirely on a finance input the model cannot see. The right briefing separates the two.

```
{
  "P_value_gt_cost": 0.8886666666666667,
  "expected_value": 52.18922582472492,
  "expected_net_value": 2.1892258247249217,
  "expected_roi": 0.04378451649449843,
  "value_lo": 49.25088852963619,
  "value_hi": 55.16479450200625,
  "cost": 50.0,
  "decision": "TEST FURTHER"
}
```

break-even margin = $1/\text{iROAS} = 33.5\%$; the campaign clears $P(\text{profitable}) \geq 0.90$ only for margins $\geq 35.5\%$



1.6 7 · Caveats

- **The estimate travels; the decision might not.** iROAS ~ 3 is for *this* creative, *this* spend level, *these* 50 geos, in Feb–Mar. A national rollout extrapolates on all four axes — diminishing returns alone (nb06’s saturation curves) argue the marginal iROAS at 10x spend is lower.

- **Profitability hinges on an input outside the data.** The margin sweep is the honest deliverable; a single “GO” at 35% margin overstates what the sales panel can know.
- **Spillover is the surviving assumption.** Randomization can’t stop Denver’s ads reaching Boulder’s shoppers; if spend leaks into control geos, both the anchor and SC are biased *downward* (control geos absorb some lift), making our estimate mildly conservative.
- **Interval optimism.** 07’s multi-seed check showed SC’s total-lift band under-covers even when the point is right. The agreement with the anchor’s *point* is earned; the band’s exact width is not.
- **75 interpolated geo-days** (0.8%, small geos, mostly pre-period) — rerun dropping the 17 gappy geos moves nothing material (they carry ~2% of donor weight combined).
- **One experiment.** This is a single draw of the world, and we’ve used its randomization once; the multi-seed calibration story lives in 07 where worlds are free.

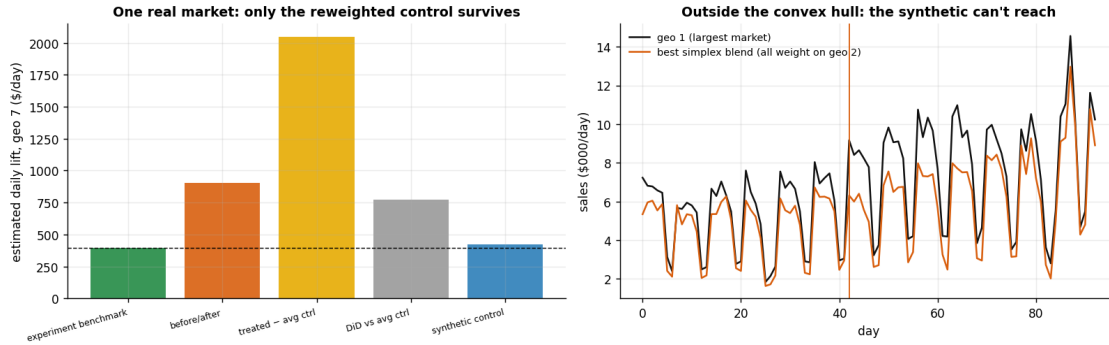
2 Depth A · The naive estimators, on a single real market

In Step 5 the naive comparisons would have looked *fine* — with 50 randomized geos aggregated into a cell, treated-minus-control is unbiased by design; that is precisely why companies run geo experiments. The moment the question becomes 07’s — **one** treated market, no coin flip — the shortcuts break on real data exactly as they broke in simulation. We replay the bake-off pretending only **geo 7** (a mid-size market, ~\$1.8k/day) ran the campaign, grading against the benchmark the full experiment implies for it (its share of spend x the anchor iROAS => ~ **+\$394/day**):

- **before/after** — books the season: sales everywhere rose into spring;
- **treated — average control** — books the *level* gap: geo 7 is ~3x the average control’s size;
- **DiD vs average control** — fixes the level, keeps the wrong *sensitivity*: a big geo swings more per unit of common shock than the average-of-50, so shared waves don’t cancel;
- **synthetic control** — reweights donors to match geo 7’s own scale and rhythm.

Then the failure 07 could only warn about: **geo 1 is the largest market in the country** — bigger than *every* control geo, hence **outside the convex hull**. The simplex does its honest best (all weight on geo 2, the largest donor), the pre-fit is visibly bad, and the **pre-fit gate FAILs** — the method itself tells you not to trust its number. That self-diagnosis is worth more than the estimate.

```
{
  "experiment benchmark": 394,
  "before/after": 904,
  "treated \u2212 avg ctrl": 2049,
  "DiD vs avg ctrl": 771,
  "synthetic control": 423
}
geo 7 gate: pre-RMSE $122 vs |daily lift|/3 = $141 -> PASS    (weights spread
over ~5 donors)
geo 1 gate: pre-RMSE $868 vs |daily lift|/3 = $665 -> FAIL    (effective donors
1.0: all weight on geo 2; implied iROAS 5.1 vs the experiment's 2.94 - inflated
~75%)
```



Read-out. Against geo 7's $\sim +\$394/\text{day}$ benchmark: **before/after** lands at **$+\$904$** (2.3x too big — it credits the campaign with spring), **treated — average control** at **$+\$2,049$** (5x — it books geo 7's size as lift), **DiD** at **$+\$771$** (2x — level fixed, sensitivity still wrong: a $\$1.8\text{k}/\text{day}$ market swings harder per common shock than the $\$0.6\text{k}/\text{day}$ average control), and **synthetic control** at **$+\$423$** , **within $\sim 7\%$** , with the gate PASSing (pre-RMSE $\$122$ vs the $\$141$ bar). Same data, four estimates, one survivor — 07's bake-off, no simulator required. And geo 1 is the counterexample that keeps us honest: the country's largest market has no counterfactual among smaller donors, the gate FAILs loudly (pre-RMSE $\$868$ vs a $\$665$ bar), and the estimate it would have produced — iROAS ~ 5.1 vs the experiment's 2.94 — is exactly the number an unscrupulous deck would have shipped. **When the treated unit escapes the donor hull, aggregate or walk away.**

3 Depth B · Placebo inference — is the gap statistically real?

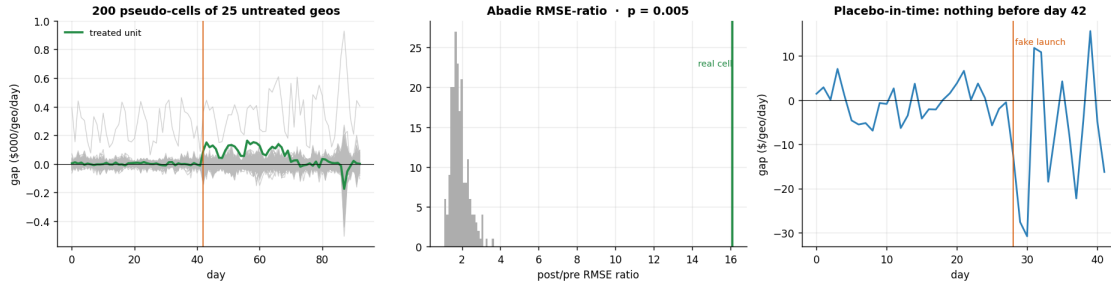
Same inferential logic as 07, adapted to the design. With the treated unit being a *cell of 25-geo scale means*, the fair placebo is not a single geo but a **pseudo-cell**: draw 25 of the 50 *control* geos at random, pretend they were a campaign cell, fit them a synthetic control from the remaining 25, and record the post-window “gap.” None of these bundles saw a single ad dollar, so their gaps map out pure noise — **the permutation distribution the design itself generates**. Two test statistics, as in 07: the raw mean post-gap, and Abadie's scale-free **post/pre RMSE ratio** (which doesn't reward bundles that simply fit badly everywhere). We also rerun 07's **placebo-in-time** — a fake launch 14 days before the real one, inside the pre-period.

This permutation test is the *calibrated* significance statement for this notebook — the posterior band is the optimistic one (07's multi-seed lesson), the rank among placebos is honest by construction.

pseudo-cell permutation: real gap $\$+105/\text{geo}/\text{day}$ vs placebo 95% range $[-17, +21]$
 $\rightarrow p = 0.010$

RMSE-ratio test: real cell 16.1 vs placebo median 1.78 $\rightarrow p = 0.005$ (no placebo bundle comes close: the real cell ranks 1st of all 201 units)

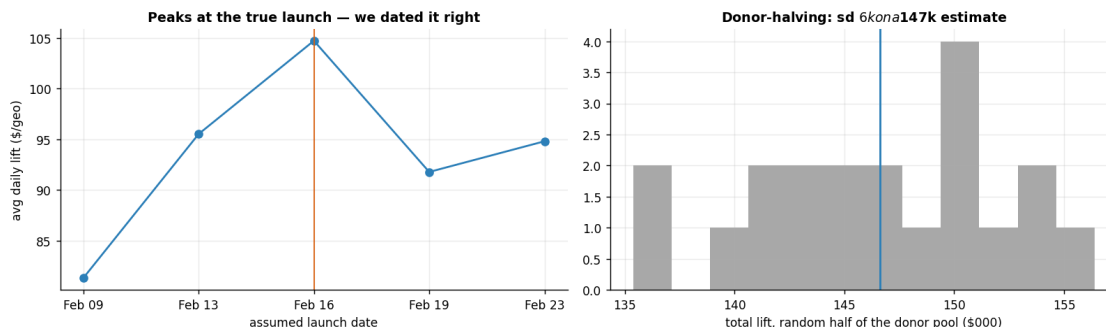
placebo-in-time: fake-launch gap $\$-8/\text{geo}/\text{day}$ - 7% of the real $\$+107$ effect and the wrong sign (pre-fit RMSE $\$4$) - no anticipation, no invented lift



4 Depth C · The dollar decision & sensitivity

Three stress tests on the number a CMO would repeat in a board meeting, then the verdict. **Launch-date sweep:** as in 07, mis-dating the campaign folds no-effect days into the window and dilutes the average — the estimate should peak *at* the true launch, and the peak is itself a diagnostic that we dated it right. **Donor-pool halving:** refit on random halves of the donor pool; a result that owns its donor pool shouldn't lurch. **The margin x iROAS surface** is already in Step 6; here we print the summary a decision memo needs.

```
{
  "total_incremental_sales_k": 149.1,
  "ci90_k": [
    140.7,
    157.6
  ],
  "iroas": 2.98,
  "did_anchor_k": 147.2,
  "did_anchor_se_k": 41.9,
  "pre_rmse_gate": "PASS",
  "p_pseudo_gap": 0.01,
  "p_rmse_ratio": 0.005,
  "placebo_in_time_gap": -7.8,
  "cooldown_gap_per_day": -2.5,
  "break_even_margin": 0.335,
  "P_profitable_at_35pct_margin": 0.8886666666666667,
  "decision_at_35pct": "TEST FURTHER"
}
```



4.0.1 The one-paragraph decision

The campaign **worked**, and we can prove it to experimental standard: the synthetic control tracks the treated cell to within $\sim \$10/\text{geo}/\text{day}$ before launch (gate PASS), lands on the randomized matched-pairs answer (\$149k vs \$147k of incremental sales; iROAS 2.98 vs 2.94), is the most extreme unit on both placebo permutation tests ($p \sim 0.005\text{--}0.010$), shows nothing material at a fake launch date ($-\$8/\text{geo}/\text{day}$, 7% of the real effect, wrong sign), survives leave-one-out and donor-halving, and — the real-data bonus no simulator provides — **switches off within days of the spend stopping**, which both certifies the attribution and denies any carryover annuity. What the data *cannot* settle is profitability: \$2.98 of revenue per ad dollar clears cost only above a **33.5% gross margin**, and reaches 90% confidence only above $\sim 35.5\%$. So the briefing is two sentences, not one: *the lift is real and precisely measured; whether to scale is a margin question* — GO for the high-margin lines, NO-GO below one-in-three margins, and any national rollout should expect the marginal iROAS to sag below this test's 2.98 (saturation, notebook 06) — so re-test at scale rather than extrapolate.